Assignment 2

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Course : Essential of Data Science

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```
[15]
from google.colab import files
uploaded = files.upload()
```

```
Choose files paper_reviews.csv
paper_reviews.csv(text/csv) - 110923 bytes, last modified: 04/05/2026 - 100% done
Saving paper_reviews.csv to paper_reviews (1).csv
```

```
[16]
import pandas as pd
import numpy as np

df = pd.read_csv('paper_reviews.csv')
print(df.shape)
df.head()
```

```
(500, 11)
```

	paper_id	reviewer_id	review_decision	confidence	novelty_score	weak_point	review_text	submission_date	venue	reviewer_location	retweet_count
0	P_1070	R_1031	reject	0.8949	3.0	Reproducibility Concerns	Major revision required. The methodology secti...	2024-01-10 00:00:00	NeurIPS	United Kingdom	9
1	P_1139	R_1011	accept	0.8918	8.0	NaN	Major revision required. The methodology secti...	2024-01-10 03:07:34	NeurIPS	China	7
2	P_1023	R_1027	reject	0.8805	2.0	Insufficient Experiments	The paper presents a novel approach to the pro...	2024-01-10 06:15:09	EMNLP	India	1
3	P_1143	R_1025	accept	0.9235	10.0	NaN	The paper presents a novel approach to the pro...	2024-01-10 09:22:43	EMNLP	United Kingdom	12
4	P_1056	R_1057	reject	0.8000	3.0	Weak Literature Review	Excellent work with clear motivation and stron...	2024-01-10 12:30:18	NeurIPS	United States	11

Next steps: Generate code with df | New interactive sheet

Problem 1:

1. Problem: Count total reviews for each review_decision.

```
[17]
df['review_decision'].value_counts()
```

review_decision	count
reject	248
accept	145
revision	107

dtype: int64

Problem 2:

2. Problem: Find total reviews submitted per paper_id.

[18] 0s

```
df.groupby('paper_id').size()
```

paper_id	0
P_1000	2
P_1002	2
P_1003	1
P_1004	5
P_1005	2
...	...
P_1144	6
P_1146	4
P_1147	1
P_1148	2
P_1149	2

140 rows × 1 columns

dtype: int64

Problem 3:

3. Problem: Identify reviewer with highest number of reject decisions.

[19] 0s

```
df[df['review_decision']=='reject']['reviewer_id'].value_counts().idxmax()
```

```
'R_1060'
```

Problem 4:

4. Problem: Compute average confidence per review_decision.

[20] 0s

```
df.groupby('review_decision')['confidence'].mean()
```

review_decision	confidence
accept	0.904894
reject	0.825357
revision	0.766066

dtype: float64

Problem 5:

5. Problem: Find most frequent weak_point category in reviews.

[21]
0s

```
df['weak_point'].value_counts().idxmax()
```

```
'Insufficient Experiments'
```

Problem 6:

6. Problem: Compute mean novelty_score across all reviews.

[22]
0s

```
np.mean(df['novelty_score'])
# or: df['novelty_score'].mean()
```

```
np.float64(4.7289915966386555)
```

Problem 7:

7. Problem: Count reviews submitted per date using submission_date.

[23]
0s

```
df['date'] = pd.to_datetime(df['submission_date']).dt.date
df.groupby('date').size()
```

date	0
2024-01-10	8
2024-01-11	8
2024-01-12	8
2024-01-13	7
2024-01-14	8
...	...
2024-03-11	7
2024-03-12	8
2024-03-13	8
2024-03-14	7
2024-03-15	1

66 rows × 1 columns

dtype: int64

Problem 8:

8. Problem: Find venue with highest proportion of accepted papers.

[24]
0s

```
df.groupby('venue')['review_decision'].apply(lambda x: (x=='accept').mean())
```

venue	review_decision
CVPR	0.252747
EMNLP	0.302752
ICLR	0.281250
ICML	0.317308
NeurIPS	0.290000

dtype: float64

Problem 9:

9. Problem: Identify review with maximum length from review_text.

[25]
0s

```
df.loc[df['review_text'].str.len().idxmax()]
```

	448
paper_id	P_1118
reviewer_id	R_1069
review_decision	revision
confidence	0.7135
novelty_score	6.0
weak_point	Unclear Methodology
review_text	The paper presents a novel approach to the pro...
submission_date	2024-03-08 08:33:40
venue	CVPR
reviewer_location	United States
retweet_count	12
date	2024-03-08

dtype: object

Problem 10:

10. Problem: Compare average review_text length across review_decision.

[26] 0s

```
df['length'] = df['review_text'].str.len()
df.groupby('review_decision')['length'].mean()
```

review_decision	length
accept	130.462069
reject	137.036290
revision	133.691589

dtype: float64

Problem 11:

11. Problem: Find busiest review submission hour using submission_date.

[27] 0s

```
df['hour'] = pd.to_datetime(df['submission_date']).dt.hour
df['hour'].value_counts().idxmax()
```

```
np.int32(0)
```

Problem 12:

12. Problem: Find venue with highest total novelty_score sum.

[28] 0s

```
df.groupby('venue')['novelty_score'].sum().idxmax()
```

```
'EMNLP'
```

Problem 13:

13. Problem: Count revision-decision reviews venue-wise.

[29] 0s

```
df[df['review_decision']=='revision'].groupby('venue').size()
```

venue	0
CVPR	25
EMNLP	21
ICLR	16
ICML	25
NeurIPS	20

dtype: int64

Problem 14:

14. Problem: Detect null values in all columns.

[30]
0s

```
df.isnull().sum()
```

	0
paper_id	0
reviewer_id	0
review_decision	0
confidence	0
novelty_score	24
weak_point	145
review_text	0
submission_date	0
venue	0
reviewer_location	0
retweet_count	0
date	0
length	0
hour	0

dtype: int64

Problem 15:

15. Problem: Find reviewer (reviewer_id) with highest review count.

[31]
0s

```
df['reviewer_id'].value_counts().idxmax()
```

```
'R_1008'
```

Problem 16:

16. Problem: Find most common weak_point for each venue.

[32]
0s

```
df[df['review_decision']=='reject'] \
  .groupby(['venue','weak_point']).size().unstack().idxmax(axis=1)
```

venue	0
CVPR	Insufficient Experiments
EMNLP	Unclear Methodology
ICLR	Insufficient Experiments
ICML	Insufficient Experiments
NeurIPS	Insufficient Experiments

dtype: object

Problem 17:

17. Problem: Calculate mean, median, std of novelty_score.

[33]
0s

```
np.mean(df['novelty_score']), np.median(df['novelty_score']), np.std(df['novelty_score'])
```

```
(np.float64(4.7289915966386555), np.float64(nan), 2.5659535207243342)
```

Problem 18:

18. Problem: Count reviews mentioning "methodology" in review_text.

[34]
0s

```
df['review_text'].str.contains('methodology', case=False).sum()
```

```
np.int64(140)
```

Problem 19:

19. Problem: Find most common reviewer_location.

[35]
0s

```
df['reviewer_location'].value_counts().head(5)
```

reviewer_location	count
United States	148
China	105
United Kingdom	65
Canada	51
India	39

dtype: int64

Problem 20:

20. Problem: Identify outliers in novelty_score using IQR.

[36]
0s

```
Q1 = np.percentile(df['novelty_score'].dropna(), 25)
Q3 = np.percentile(df['novelty_score'].dropna(), 75)
IQR = Q3 - Q1
outliers = df[(df['novelty_score'] < Q1 - 1.5*IQR) |
              (df['novelty_score'] > Q3 + 1.5*IQR)]
print(outliers[['paper_id','reviewer_id','novelty_score']])
```

```
Empty DataFrame
Columns: [paper_id, reviewer_id, novelty_score]
Index: []
```